BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

LESSON 316 · Paper 1 extended response challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does evaluate mean?

Q2. What is a counterargument?

Q3. Why mention the scenario?

Q4. Half adder A=1 B=1. Give sum and carry.

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I Do · Read new knowledge

Key vocabulary

evaluation — Supported comparison and decision.

trade-off — Improvement with an associated cost.

context — Details of the specified use case.

justification — Evidence supporting a decision.

New learning

Step 1 — Identify scenario-specific objectives and two valid alternatives.

Step 2 — Explain mechanisms and trade-offs with accurate technical details.

Step 3 — Conclude with judgement grounded in costs, users, performance and security.

Worked example

Evaluate SSD versus HDD for a mobile video editor.

1. Explain the worked method — A strong answer explains mechanical access, portability and the need for backups.

We Do · Practise with guidance

Evaluate cloud-based coursework storage against USB-only storage for a school.

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Cloud: convenient multi-device access and managed permissions. USB: offline
  access, but loss/damage risk.
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Explain the worked method

Match security, home connectivity and backup provision to the school context.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Improve 'the cloud is better' for a school, explaining a benefit, limitation and judgement. [6]

Question 2

Evaluate GPU versus multicore CPU for video analysis and general tasks. [9]

Question 3

Evaluate nightly local-only backups versus secure independent off-site frequent backups for school records. [9]

Exit Ticket · Answer before checking

Many disconnected technical terms guarantee full extended-response marks.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Make a reasoned judgement, with relevant supporting evidence.

Q2. A relevant limitation to the preferred choice.

Q3. It decides which advantages and limitations matter.

Q4. Sum0 from XOR; carry1 from AND.

You Do pages · Independent questions

Q1. Multi-device access and shared controls [2]; outage/privacy limitation [2]; justified view based on school connectivity/backup [2].

Q2. Indicative content: GPU parallel throughput, CPU suitability for serial/branch-heavy operations, transfer overhead, cost and energy. Practice marking (authored guidance, not an OCR mark scheme): 7–9 = accurate, sustained evaluation of both options, applied to this scenario with a supported judgement; 4–6 = relevant comparison with some development and a partly justified decision; 1–3 = basic statements, limited context or unsupported decision; 0 = no creditworthy response. Use best-fit and credit equivalent valid arguments.
Model evaluation: For a large batch of independent video frames, a GPU can perform many similar pixel operations simultaneously, so it may substantially improve throughput. A multicore CPU is more flexible for the control logic, branching and varied tasks used elsewhere in the application. Moving data between CPU and GPU memory can add latency, so a small job may finish faster without GPU offloading. Power consumption, hardware cost and development effort also matter. For large-scale video analysis I would use GPU acceleration for suitable parallel stages, retaining CPU cores for orchestration and general tasks. The decision should be tested on realistic workloads rather than inferred from core counts alone.

Q3. Indicative content: recovery point objective, local-site failure/ransomware, backup frequency, off-site independence, encryption, access controls, restore tests and cost. Practice marking (authored guidance, not an OCR mark scheme): 7–9 = accurate, sustained evaluation of both options, applied to this scenario with a supported judgement; 4–6 = relevant comparison with some development and a partly justified decision; 1–3 = basic statements, limited context or unsupported decision; 0 = no creditworthy response. Use best-fit and credit equivalent valid arguments.
Model evaluation: A nightly local backup is inexpensive and may restore an accidentally deleted file quickly, but a failure just before the next backup could lose almost a day's work. Fire, theft or ransomware could also affect both live files and local copies. Secure, independently controlled off-site backups taken more frequently reduce the likely loss of recent records and protect against a building-level incident. However, they bring ongoing cost, connectivity and data-protection responsibilities. I would recommend frequent encrypted off-site or immutable backups alongside a convenient local copy, with restricted access and regular restore tests. The school should choose frequency against an agreed maximum acceptable data loss and recovery time.

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Exit Ticket · Model answer

False. Relevant explanation and contextual evaluation matter.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: